

Human Forms of Reasoning

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1. Introduction

This paper is about kinds of reasoning that are unique to humans. Distinguishing between direct (e.g. *Modus Ponens*, *Conjunction Elimination*) and indirect (e.g. *Hypothetical Reasoning*, *Reductio ad Absurdum*) forms of reasoning, we suggest that humans and only humans are capable of the latter. We provide evidence that non-human great apes (from here on: apes) can do reasoning, but the evidence we have of their reasoning is compatible with it being direct only. Evidence for employment of indirect forms of reasoning in human cognition comes from natural language semantics/pragmatics and from research on psychological defense mechanisms. Indirect forms of reasoning require access to earlier stages of computation, i.e. access to the derivational history of the reasoning in question, which relies on extended involvement of the procedural memory. We tentatively suggest that the uniqueness of humans in various forms of reasoning is related to a boost in the procedural memory within the evolution of *homo sapiens*.

Explicating the unique nature of human reasoning requires some understanding of different types of reasoning in logic, where the central concept is that of validity. Logic is about distinguishing those arguments that are valid from those that are not. Consider the argument in (1), which is an example of a non-valid argument.

(1) Premise 1. If John is at home, his lights are on.

Premise 2. John's lights are on.

Conclusion: John is at home.

The conclusion might, of course, be true, but the argument is not valid. It could be that John is the kind of person who leaves the lights on all the time, whether he is at home or not. Then it will be true that if he is at home, his lights are on and that his lights are on at the moment, but it need not be the case that John is at home at this moment. Whenever an argument involves a conclusion which can be false while all the premises are true, we say that the argument is not valid. Logic is concerned with general principles that make valid arguments valid and invalid arguments invalid.

Turning this invalid argument above into a valid one is not that difficult. We swap the order of Premise 2 with Conclusion to obtain (2), which is a valid argument

(2) Premise 1. If John is at home, his lights are on.

Premise 2. John is at home.

Conclusion: John's lights are on.

There is some kind of indefeasibility about the argument in (2). Once you accept the premises as true, you are trapped (or liberated, as it may be); there is nothing you can do but accept that the conclusion is true. What makes this argument valid? It is nothing personal about John, or about him being at home. Neither is it about the lights being on at the relevant moment. We can change every one of these parameters and what we have is still a valid argument.

(3) Premise 1. If Mary is at school, then her television is off.

Premise 2. Mary is at school.

Conclusion: Mary's television is off.

The validity of the argument is not about the meaning of the pieces that make up the sentences in the premises and in the conclusion. It is about the general form of these premises and the conclusion. In this sense, validity is a form-related concept. The arguments in (2) and (3) are valid because they instantiate the Modus Ponens schema (also known as *Conditional Elimination*, or " $\rightarrow E$ ") shown in (4), where " ϕ " and " ψ " are metavariables over simple and complex sentences.

(4) Modus Ponens

$$\begin{array}{ccc} \phi \rightarrow \psi & \phi & \\ \hline & \psi & \rightarrow E \end{array}$$

Modus Ponens is a local rule in that in applying it, we look at the previous two propositions and only them. Such forms of reasoning are called *direct* reasoning, meaning that the conclusion follows *directly* from the last (two) premises. This is not the only kind of reasoning representable in logic, however. Suppose that I am curious as to whether John's cat is happy. Now, I know that

if John's lights are on then his cat is happy and I know, moreover, that if John is at home then his lights are on. From these two premises I conclude that John's cat is happy *provided that he is at home*. In this style of argument, premises are evaluated in the context of a supposition, which gets to be discharged at the level of conclusion.

(5) Supposition: John is at home

 Premise 1. If John is at home, then his lights are on.

 First Interim Conclusion. John's lights are on.

 Premise 2. If John's lights are on, then his cat is happy.

 Second Interim Conclusion: John's cat is happy.

 Conclusion: If John is at home, then his cat is happy.

The form of this reasoning in (5) can be represented in two steps. There are two rules of Modus Ponens that must be applied before we can derive the overall conclusion. The suppositional status of the sentence "John is at home" is indicated with the superscript "1".

$$\begin{array}{lcl}
 (6) & p^1 & p \rightarrow q \\
 & \hline
 & q & q \rightarrow r \\
 & \hline
 & & r
 \end{array}$$

We will now eliminate the suppositional status of p by inserting it as the antecedent of a conditional at the lowest level in the derivation. In this form of reasoning, we conditionalize an interim conclusion to a supposition, which is why such a reasoning is sometimes called *Hypothetical Reasoning*. The symbol " $\rightarrow I^1$ " should be read as the rule of conditional introduction targeting the premise indexed with 1. The elimination of a suppositional premise is indicated with a strikethrough over the premise.

$$\begin{array}{c}
 (7) \quad \frac{p^{\dot{}} \quad p \rightarrow q}{\quad} \\
 \frac{q \quad q \rightarrow r}{r} \\
 \frac{\quad}{p \rightarrow r} \rightarrow I^1
 \end{array}$$

The general syntactic schema of the rule of Conditional Introduction is given in (8). Observe that, unlike Modus Ponens, this rule requires access to earlier stages in the argument, as signaled by the dots. In other words, the conclusion is *not* a function of the last steps in the derivation. This is why this form of reasoning is called *indirect*: Conditional Introduction is an indirect form of reasoning. In this proof, the sentence p (indexed with j to express its suppositional status) is said to be *discharged* by the application of the rule of Conditional Introduction targeting it.

(8) Conditional Introduction

$$\begin{array}{c}
 \phi^j \\
 \vdots \\
 \psi \\
 \hline
 \phi \rightarrow \psi \rightarrow I^j
 \end{array}$$

The kind of rules we find in (4) and (8) are called Gentzen Rules named after Gerhard Gentzen, who introduced them (Gentzen, 1934). However, the distinction between direct and indirect forms of reasoning is not a contingency of the Gentzen schema. A similar distinction is observed in other proof systems. Inspired by the work of Jaśkowski (1934), Fitch (1952) introduced an alternative proof system in which a reasoning involving Modus Ponens is expressible using a single proof line.

(9) Modus Ponens in Fitch Deduction

1	$p \rightarrow q$	Given
2	p	Given
3	q	Modus Ponens from 1 & 2

The implementation of the reasoning we have seen in (7) within Fitch Deduction involves a second, embedded proof, a sub-proof embedded inside the main proof. This sub-proof is indicated with a second line embedded inside the main proof line

(10) Conditional Introduction in Fitch Deduction

1	$p \rightarrow q$	Given
2	$q \rightarrow r$	Given
3	p	Assumption
4	q	Modus Ponens from 1 & 3
5	r	Modus Ponens from 2 & 4
6	$p \rightarrow r$	Conditional Introduction from 3 & 5

In Gentzen Rules, Conditional Introduction makes reference to an earlier statement in the proof, which is marked as a supposition. In Fitch Deduction, on the other hand, Condition Introduction refers to a sub-derivation and its function is to eliminate the assumption of this subproof in a way that enables reasoning back in the main proof line. Both formalisms make it clear that Conditional Introduction involves a type of complexity that we do not find in Conditional Elimination (i.e. Modus Ponens). In other words, the complexity of indirect forms of reasoning as opposed to direct forms of reasoning is constant across these two proof formalisms. Throughout the paper, we take Gentzen Rules as our model of the proof system.

Conditional Introduction is not the only form of indirect reasoning. Suppose that you suspect that the set of real numbers is a countable set. What this means is that it should be possible to find a bijective function from the set of natural numbers, which is countable, to the set of real numbers. Applying the diagonalization argument devised by Georg Cantor, you come to see that no such bijection can be found. The assumption that the set of real numbers is countable has led you into a contradictory state, from which you conclude, accurately, that the set of Reals is not countable. (This is a type of reasoning that is generally known as *Reductio Ad Absurdum*). Underlying the reasoning in this example is the rule of Negation Introduction “ $\neg I$ ” targeting an assumption (e.g. “the set of Reals is countable”) made in the earlier stage in the derivation, which is an indirect form of reasoning. This rule has the following schema, where we use “ $\perp$ ” to express contradiction.

(11) Negation Introduction

$$\begin{array}{c}
 \varphi^j \\
 \vdots \\
 \perp \\
 \hline
 \neg I^j \\
 \neg \varphi
 \end{array}$$

We can derive the rule of Negation Elimination by postulating an entailment from “ $\neg\neg\varphi$ ” to “ φ ”. What we do is choose “ $(\neg\psi)^j$ ” as the input for the rule of Negation Introduction, as a result of which we derive “ $\neg\neg\psi$ ”, which entails “ ψ ” by our assumption. This is the rule of Negation Elimination. In other words, Negation Elimination runs Negation Introduction as a subroutine, which is why we consider it to be an indirect form of reasoning.¹

Let us also note that Conditional Elimination is not the only direct inference rule in logic. We can obtain a conjunction by combining conjuncts and we also can eliminate conjuncts from a conjunction. These are the rules of Conjunction Introduction and Conjunction Elimination respectively. Note that Conjunction Elimination can target any of the conjuncts.

(12) a. Conjunction Introduction

$$\begin{array}{c}
 \varphi \qquad \psi \\
 \hline
 \varphi \wedge \psi \quad \wedge I
 \end{array}$$

b. Conjunction Elimination

$$\begin{array}{c}
 \varphi \wedge \psi \\
 \hline
 \{\varphi, \psi\} \quad \wedge E
 \end{array}$$

We now have rules of logic that insert or eliminate operators and connectives. Some of the rules are direct, meaning they only involve the result of the last steps of derivation. Others are indirect, meaning they target assumptions made in the earlier stages of the derivation. We can summarize our observations as follows:²

¹ As a matter of fact, there is also a direct way of expressing the rule of Negation Elimination. Given some formula of the form “ $\neg\varphi$ ”, we match it with a formula of the form “ φ ” to obtain a contradiction and, by principle of explosion, from contradiction, any formula, including “ φ ”, follows, which is how we end up eliminating the negation operator. Paraconsistent systems of logic have been developed to avoid the principle of explosion as a legitimate property of logic (Priest, 2002). In fact, as we will see at the end of Section 3, human mind is very much disturbed by contradictory states, which leads to cognitive dissonance (Festinger, 1957). We find the indirect way in which negation is eliminated in the main text more natural and easier to work with, although it must be said that nothing crucial hinges on this assumption. We will not make use of the rule of Negation Elimination in the rest of the paper.

² We do not include rules introducing and eliminating disjunction in our list due to their unnaturalness. Given some formula “ φ ”, we can derive “ $\varphi \vee \psi$ ” by a weakening rule of logic. It is, however, difficult to see how such a rule would be of any use to an animal. In order to eliminate disjunction in a formula of the form “ $\varphi \vee \psi$ ”, we turn each disjunction into a suppositional state (e.g. φ^i and ψ^i) and derive a formulae χ from each of them so that we can

Direct Forms of Reasoning:

Conjunction Introduction, Conjunction Elimination, Conditional Elimination

Indirect Forms of Reasoning:

Conditional Introduction, Negation Introduction, Negation Elimination

The central claim of this paper is that *only humans mentally perform indirect forms of reasoning*, the type of reasoning that requires access to earlier stages of the computation. We suggest, tentatively, that indirect reasoning requires extended involvement of the procedural memory, i.e. a boosted procedural memory, which is possibly a uniquely *homo sapiens* property (see Section 4 for a detailed discussion of this point).

Before we discuss direct forms of reasoning in primate cognition, it is important to distinguish between simple and complex schemas of inference. The rules we have given above are basic forms of reasoning in that they involve addition and elimination of connectives. It is also possible to summarize complex interactions between these basic rules as valid schemas. For instance, the kind of proofs that we have seen in (7) and (10) can be summarized as in (13), which is called Transitive Inference.

(13) Transitive Inference

$$\begin{array}{c} \varphi \rightarrow \psi \quad \psi \rightarrow \chi \\ \hline \varphi \rightarrow \chi \end{array}$$

Looking at the schema in (13), we may come to think that it instantiates a direct form of reasoning given that the consequence appears to be a direct consequence of the last two premises. This impression is misleading, however. It is important to note that what we call Transitive Inference is obtained by the application of the rule of Conditional Introduction subsequent to the two instances of Conditional Elimination. In other words, the representation of Transitive Inference in (13) is not indicative of its internal complexity. Therefore, as far as this paper is

replace “ $\varphi \vee \psi$ ” with “ χ ”, thus eliminating disjunction. However, entertaining double suppositional states does seem to be cognitively costly. Moreover, many times, we cannot distinguish between a disjunctive sentence of the form “ $\varphi \vee \psi$ ” from a conjunction of possibilities, $\text{Poss}(\varphi) \wedge \text{Poss}(\psi)$, in an experimental setting. In the next section, we will provide an example of how disjunction can be re-interpreted as the conjunction of possibilities in the context of an experiment on the cognition of uncertainties by great apes (Engelmann et al., 2021).

concerned, Transitive Inference is a complex form of indirect reasoning since it involves direct and indirect forms of reasoning as its constituents. In a sense, Transitive Inference is a summary of some complex interactions between basic forms of direct and indirect reasoning. This is an important consideration to keep in mind in evaluating the internal complexity of an argument schema.

2. Direct Forms of Reasoning in Ape Cognition

In this section, starting with Conjunction Introduction, we provide evidence that apes can mentally implement direct forms of reasoning. We must, first, distinguish between conjunctive thoughts and conjunctive reasoning. It is possible that conjunctive thoughts are widespread in ape cognition, and more generally in animal cognition, in representing an external object, event or situation. Hurford (2007, p. 114), for instance, suggests that “part of what a baboon perceives when it contemplates a scene containing a crouching lion and a rock” can be represented as “ $LION_{baboon}(x) \ \& \ CROUCH_{baboon}(x) \ \& \ ROCK_{baboon}(y)$ ” where the index “baboon” is intended to indicate that these concepts are relativized to the baboon cognition, i.e. whatever the baboon cognizes when she cognizes some animal as a lion or as crouching, and the variables x and y stand for “the mental indices of these objects in the baboon’s brain as it tracks them (probably realized as activity in its posterior parietal cortex)”. An animal that can track objects and register their properties can be attributed “a (limited) capacity for logical conjunction”, limited presumably by the constraints on the number of objects it can track (see the discussion in Hurford, 2007, pp. 90 – 96). Similarly, Tomasello (2014) suggests that a chimpanzee approaching a banana tree with the intention of foraging attends to a set of situations that are relevant to her decision making. These situations may include: (a) that there are many bananas in the tree, (b) that the bananas are ripe, (c) that the bananas are reachable by climbing, (d) that no predators are nearby etc ... and it is a conjunction of these situations that explain the foraging behavior of the chimpanzee. These sub-situations that are relevant to implementing a specific behavior are “all present in a single visual image.” (p. 11).

Conjunctive reasoning goes beyond logical (here, conjunctive) analysis of the stimulus. Conjunctive reasoning should involve integrating distinct types of information from the environment that are causally relevant to the execution of behavior. Experimental evidence for conjunctive reasoning in great apes can be found in the future-oriented tool manipulation of a

zoo-housed Sumatran orangutan by the name of Riau (Mulcahy, 2018). In an enrichment task involving the use of a long and heavy tool to rake-in the rewards that were out of his reach, Riau was observed to spontaneously hang up the tool into gaps of the mesh fence surrounding his enclosure, presumably to avoid the cost of re-carrying the tool from the ground when it would later be needed. Mulcahy (2018) ran two experiments on Riau's tool-oriented behavior, one to test the hypothesis that this behavior is contingent on the *future* availability of the reward and a second experiment to show that the size and/or shape of the tool is a decisive factor in deciding to implement this behavior. The first experiment showed that in 59 out of 60 trials Riau wedged the tool into the narrow gaps of the fence when he could see that the tool would be needed in a short while in order to retrieve more of the reward. Moreover, he almost never exhibited this tool-oriented behavior when he could see no future use for the tool (1/12). The second experiment showed that when given a smaller and lighter tool, he never exhibited this securing behavior.³ The conjunctive reasoning underlying this behavior can be represented as in (14), where *FUT* is a temporal operator that locates the event or situation denoted by the sentence at a time later than *now*:

(14) BEHAVIOR: hangs up the tool

REASONING UNDERLYING THE BEHAVIOR:

$$\begin{array}{ccc}
 \text{large(stick)} & & \text{FUT(available(food))} \\
 & \text{—————} & \wedge I \\
 & & \text{large(stick) } \wedge \text{ FUT(available(food))}
 \end{array}$$

Some authors deny non-human animals the capacity to represent sentence-like structures, which means that they also deny them the capacity for logical reasoning, the kind of reasoning that is based on sentential operators. Bermúdez (2006, p. 129), for instance, claims that “an account of animal reasoning” requires that we “identify forms of reasoning at the non-linguistic level and then explain them without assuming that the animal is deploying elementary logical concepts or exploiting the internal structure of a thought”, where logical concepts include

³ The second experiment also involved a control where Riau manipulated the large tool. In all cases where there was future use for the tool, Riau wedged it inside the gaps of the fence (30/30). In those cases where there was no more reward to be retrieved, he did not exhibit this securing behavior (0/6). This strenghtens the conclusions of the first experiment.

conjunction, disjunction, negation, quantification etc. We have already seen some experimental evidence that a great ape is able to mentally represent tensed thoughts. The tense operator is understood to locate an event or situation, which is the denotation of a sentence, with respect to the contextual variable *now*, i.e. it is an operator that takes sentences as input, hence the symbolization “FUT(ϕ)” given some sentence ϕ .⁴ Similarly, Engelmann et al. (2021, p. 1377) provide experimental evidence that “chimpanzee thought is not limited to what is, but also involves reasoning about what could be the case.” In their experiments, two containers were placed on the top of two platforms where a rope, which extended into the testing room, was treaded through a loop on each platform. When the chimpanzees knew where the food was, either because the containers were transparent or because they observed the experimenter put the food into the opaque container, they were more likely to pull the only end of the string that provided immediate access to the platform with the baited container. However, when the chimpanzees did not know where the food was, they were more likely to pull the string on both ends to gain access to both of the platforms at the same time. The reasoning underlying the behavior of pulling the string on both ends can be represented as “POSS(has(container1, food)) $\wedge$ POSS(has(container2, food))”, where POSS is a sentential possibility operator which says that the state of affairs denoted by the sentence that POSS takes as argument cannot be ruled out/ignored.⁵ As a matter of fact, both tense operators (e.g. past, future) and modality operators (e.g. possibility, necessity) act on sentences, which means that an animal that can mentally manipulate these operators (e.g. an animal that can conceptualize future contingencies and/or alternative possibilities) can also mentally represent sentences as units of symbolic manipulation. There is, then, no conceptual reason why such an animal should not do logical reasoning.

How about Modus Ponens? Tomasello (2016, p. 62) finds evidence for a version of this type of reasoning in the tool use of great apes, observing that they “make inferences all day every day of the Modus ponens type: Poking with a stick causes ants emerge. I poke. This causes ants

⁴ This is independent of whether tense is analyzed as a quantificational operator in the spirit of Prior (1967), where “FUT(ϕ)” is interpreted as “ $\exists t. \phi(t) \wedge \text{follows}(t, \text{now})$ ”, or as a referential operator, as in Reichenbach (1947) and Partee (1973), where “FUT(ϕ)” is interpreted as “ $\phi(t) \wedge \text{follows}(t, \text{now})$ ” with the temporal variable t getting its value directly from the assignment function g .

⁵ Engelmann et al. (2021) provides a disjunctive analysis for the reasoning underlying this behavior: *has(container1, food) $\vee$ has(container2, food)*. This does not change the gist of the argument we are making. If the great apes are capable of disjunctive thoughts, they are also capable of sentence-like representations. We opt for an analysis with conjunction of possibilities partly because we submit that, as a logical operator, disjunction is really difficult to work with. See the discussion in footnote 2.

to emerge.” Importantly, there may be an alternative way of thinking about the tool use in question. In this case, tool use may instantiate a type of *kind reasoning*: A great ape coming across a stick may entertain the thought that this is the *kind of thing* that makes ants available. In this sense, kind reasoning relates objects (the stick) to situations (the availability of ants). This reasoning should be distinguished from Modus Ponens: The conditional statement inside Modus Ponens relates situations to situations, where situations are construed as the denotation of sentences. Better evidence for Modus Ponens in ape thought would be a case in which an object leads to a situation only if some other situation also holds, a type of reasoning that we shall call *conditionalized kind reasoning*. The details of tool use by chimpanzees in Gougoulo, the Republic of Congo, provide some evidence for conditionalized kind reasoning in great apes.

Gougoulo chimpanzees make and use tools for fishing termites located (invisibly) inside an earth mound (see Byrne et al., 2013 for a detailed description). The process begins with *nest puncturing*, where the chimpanzee pushes a sturdy stick into the ground to open a tunnel that enables access to the termite colony.⁶ This is followed by *termite fishing* during which the chimpanzee straightens the brush fibers of the fishing probe each time before inserting it into the nest. The next step is *tunnel clearing*, in which the chimpanzee “reverses the orientation of a brush-tip tool and uses the blunt end to clear loosened soil from the entrance of the fishing hole.” (Byrne et al., 2013, p. 56). What is important for our purposes is that the subroutine involving termite fishing *and* tunnel clearing is repeated several times in this order. In fact, Byrne et al. (2013) provide a minute-by-minute description of the loop on this subroutine. After nest puncturing at 9:07 and termite fishing at 9:15, the following steps are taken by the chimpanzee: (9:27 tunnel clearing, 9:28 termite fishing), (9:28 tunnel clearing, 9:28 termite fishing), (9:32 tunnel clearing, 9:33 termite fishing)... The loop on this subroutine suggests that the chimpanzee is aware that the stick provides access to ants only if the brush fibers are straight and the tunnel is clear. This reasoning can be represented as in (15)

⁶ The end point of the tool is smelled or visually inspected to see if access to the colony has been achieved. The execution of such test routines provides evidence that what the chimpanzee is doing goes beyond a linear associative behavioral sequence learnt from past contingencies. The chimpanzee represents a goal state and makes changes to the world to attain it (see Byrne et al. 2013 for further discussion).

(15) REASONING UNDERLYING TERMITE-FISHING BEHAVIOR

straight(fiber) $\wedge$ clear(tunnel) $\rightarrow$ available(ant) (by the affordances of the stick)

straight(fiber) $\wedge$ clear(tunnel)

available(ant) $\rightarrow$ E

The reasoning underlying this complex behavior is an example of the conditional kind reasoning that we have hinted at above. Note that the chimpanzee behaves selectively to bring about the state of affairs denoted by each of the conjuncts inside the conjunctive antecedent “straight(fibers) $\wedge$ clear(tunnel)”⁷. The loop on the subroutine involving termite fishing and tunnel clearing is the chimpanzee trying to sustain this conjunctive state. This provides evidence that the conjunctive analysis of the conditional clause we have adopted in (15) reflects the pieces of reasoning underlying complex termite-fishing behavior more or less accurately.

It must be noted that there are some noticeable differences between the conditional statement in (15) and the material implication of formal logic. The states expressed by the two clauses inside the conditional are linked by the concept of *affordance*. In formal logic, however, there need not be any link between the two sentences joined by the material implication. This is not surprising. The material analysis of conditionals in natural languages is beset with unintuitive results (sometimes called *paradoxes of material implication*). For instance, for any two propositions p and q , either “ $p \rightarrow q$ ” holds or “ $q \rightarrow p$ ” holds, which does not feel intuitive at all. Similarly, if you want to negate the claim that if Santa Clause lives in Reykjavík, then he has no beard, you must assent that Santa Clause is a bearded person living in Reykjavík given the equivalence between “ $\neg(p \rightarrow q)$ ” and “ $p \wedge \neg q$ ” (see Egré & Rott, 2021 for a lengthy discussion of these issues). Explaining the *necessary connection* between the antecedent and the consequent is the main aim of modal logics that are intended to characterize conditionals in natural languages (Stalnaker, 1968; Kratzer, 1986 among many others). For instance, a sentence like “if an object is unsupported then it falls” is understood to mean something like “by the laws of physics, if an object is unsupported then it falls”, or more precisely as “given a possible situation s in which the

⁷ It is not suggested here that the chimpanzee in question has the human concepts of *straight*, *fiber*, *tunnel* etc. Imitating Hurford’s notation, we could index each of these concepts with the word *chimpanzee* to highlight the fact that these concepts should be relativized to chimpanzee cognition, e.g. CLEAR_{chimpanzee}(tunnel_{chimpanzee}).

laws of physics are operative and an object is unsupported, the object falls in s”. In our examples, this connection between the antecedent and the consequent is formed via the affordances of the stick used by the chimpanzee.⁸

It has been suggested that many animals are capable of transitive inferences (see Lazareva, 2012 and Gazes & Lazareva, 2021 for overview and see Vasconcelos, 2008 for a defense of explanations based on associative learning procedures). In the typical experimental procedure testing the availability of such inferences, an animal is given four pairs of five stimuli with differential reward contingencies: (A+B–), (B+C–), (C+D–), (D+E–), where “+” means the associated stimulus is rewarded and “–” indicates the absence of any reward.⁹ Once a predetermined criteria of correct performance is achieved, animals are tested on the pair (B, D), with which they have had no previous experience. Even though B and D are rewarded with identical frequency in the training, many animals tend to choose B over D. This result has sometimes been analyzed as an instance of transitive inference, where given the prominence of B over C and the prominence of C over D, the animal under test deduces the prominence of B over D and behaves accordingly.

Do such experiments really capture instances of logical transitive reasoning in animals? At the end of the last section, we have seen that Transitive Inference, in its logical instantiation, is really a schematic short-hand for two applications of Modus Ponens followed by the rule of Conditional Introduction. As we have seen, the rule of Conditional Introduction starts with a supposition that we typically do not know to be true (Hypothetical Reasoning) or perhaps we even know to be false (Counterfactual Reasoning). We, then, calculate the consequences of this supposition only to discharge it at the end of the derivation as the antecedent of a conditional statement. However, what is called Transitive Inference in the experiments described above is really about locating stimuli on an ordered line and has little to do with considering suppositional states of affairs. The relevant representations are “orderly relations among stimulus items” (Dusek & Eicherbaum, 1997, p. 7133). In other words, they are representations that involve

⁸ We will not comment on the rule of Conjunction Elimination in ape cognition, which is a weakening rule in that the conjunction is always logically stronger than its conjuncts. If an ape represents a stimulus conjunctively, as Hurford and Tomasello claim they do, then Conjunction Elimination may be operative in the capacity to attend to non-salient properties of a stimulus. Given a stimulus which is analyzed as “ $p \wedge q$ ”, where p is salient and q is not, the ape can derive q and behave in a way that is solely sensitive to q being the case.

⁹ More than three pairs are needed in the training phase so as to guarantee that the pair that is tested does not contain a stimulus that has always been rewarded or one that has never been rewarded.

objects ordered on a line, i.e. linear representations. In this sense, these inferences are more geometric than logical. We, therefore, take it that such experiments do not provide convincing evidence that non-human animals are capable of indirect forms of reasoning in the logical sense. The question, then, is: Do we have evidence that humans are capable of indirect forms of reasoning in this sense? The answer seems to be positive and this is the topic of the next section.

3. Indirect Forms of Reasoning in Human Cognition

In this section, we provide evidence for implicit uses of indirect forms of reasoning in human cognition. We find evidence for this kind of reasoning in the linguistic phenomena of presupposition projection and conditional representation as well as in the capacity for symbolic pretense in children and the psychological defense mechanism of Rationalization.

The presuppositions of a sentence are the states of affairs that must hold for the sentence to have a truth value (Strawson, 1950). For instance, to decide whether (16) is true in the actual world, we must take it for granted that the Republic of Türkiye has a prime minister.¹⁰

(16) The prime minister of Türkiye resides in Ankara.

presupposes: Türkiye has a prime minister

The presuppositional status of an expression can be ascertained by applying the *Wait-a-minute* test (von Stechow, 2004), which can target the presupposition of a sentence but gives rise to semantic anomaly, expressed by adding the hash symbol in front of the sentence, when used against the assertion of a sentence.

(17) A: The prime minister of Türkiye resides in Ankara

B: Hey, wait a minute. I never knew Türkiye had a prime minister.

B': #Hey, wait a minute. I never knew he resided in Ankara.

The definite determiner *the* in English is said to be a presupposition trigger. Other presupposition triggers in natural languages are possessive pronouns (e.g. The sentence *his guitar is made in*

¹⁰ As a matter of fact, this is false since this governmental position has been dissolved as a result of the 2017 Referendum.

Korea presupposes that he owns a guitar, perhaps a unique one), factive verbs (e.g. *know*, *remember*, *realize*), verbs of change of state (e.g. *stop* drinking, *begin* teasing), adverbials (e.g. *again*, *anymore*), focus particles (e.g. *too*, *even*) among many others.

The presupposition of a sentence typically remains intact when the sentence is negated (18a), turned into a question (18b) or used as the antecedent of a conditional (18c). All the sentences in (18) presuppose that the Republic of Türkiye has a prime minister.¹¹

- (18) a. The prime minister of Türkiye does not reside in Ankara
 b. Does the prime minister of Türkiye reside in Ankara?
 c. If the prime minister of Türkiye resides in Ankara, then where does the president reside?

This phenomenon is known as presupposition projection, where a presupposition associated with a trigger is understood to be the presupposition of the whole sentence (i.e. it projects through operators such as negation, question and conditional, so to speak). There are also cases in which a presupposition does not fully project but only partially, i.e. as a *conditional* presupposition (Karttunen, 1974 see also Heim, 1983, Schlenker, 2008, Fox 2008 a.o.).

- (19) If John is a musician, then he will bring **his** guitar to the party.

presupposes: if John is a musician, then he owns a guitar

does not presuppose: John owns a guitar.

Karttunen (1974) suggests that while the whole sentence in (19) is evaluated against a context set *c*, the consequent of the conditional is evaluated against the context set *c'* which is the result of updating of the context set *c* with the antecedent (i.e. *that John is a musician*). In other words, the sentence *he will bring his guitar to the party* is evaluated against *c'*, which is just like *c* except that the set of possible situations/worlds in which John is not a musician are eliminated. Given our contextually supplied knowledge (i.e. our contextual world knowledge) that *musicians (of John-type) have guitars*, it follows that John has a guitar, which is how the presupposition of the consequent gets to be satisfied. This idea can be implemented as an instantiation of the rule of

¹¹ In other words, all these sentences feel misguided to someone who is acquainted with the results of 2017 Referendum in Türkiye.

(20) $\forall x. \text{musician}(x) \rightarrow \exists y. \text{guitar}(y) \ \& \ \text{own}(x, y)$ [Contextual World Knowledge]

$$\frac{[\text{musician}(\text{john})]^1 \quad \text{musician}(\text{john}) \rightarrow \exists y. \text{guitar}(y) \ \& \ \text{has}(\text{john}, y)}{\exists y. \text{guitar}(y) \ \& \ \text{has}(\text{john}, y)} \rightarrow E$$

Conditional Introduction involves reasoning with a supposition. A speaker's attitude towards this supposition may vary: the speaker may consider it to be a live option about the world (Hypothetical Reasoning) or the speaker may consider it to be unlikely to be the case (Counterfactual Reasoning). This distinction has grammatical ramifications reflected in the categories of indicative vs subjunctive conditionals. Consider (21a) and (21b) from Adams (1970, see also the discussion in von Stechow, 2011)

- Knowing that Kennedy was assassinated, we are ready to accept (21a) as true. However, accepting (21b) as true requires some non-trivial assumptions about American political life. Karttunen and Peters (1979) put the distinction between indicative and subjunctive conditionals in terms of possibility with respect to what is known. In the indicative mood, a sentence of the

form “If A, then B” highlights the fact that the sentence *A* is epistemically possible. In the subjunctive mood, the same sentence puts focus on the fact that it is epistemically possible that the sentence *A* is false. In any event, by hypothesis, the ability to represent hypothetical and counterfactual scenarios and reason within such suppositional states is a capacity that enables humans to distance themselves from the actual states of affairs, a capacity enabled by the rule of Conditional Introduction. This is, indeed, a powerful mode of thinking. In response to various complaints about the reliability of suppositional reasoning, Steiner (1998, p. 226), in *After Babel*, writes: “Hypotheticals, ‘imaginaries’, conditionals, the syntax of counterfactuality and contingency may well be the generative centers of human speech.”

It is likely that the rule of Conditional Introduction has something to do with symbolic forms of *pretense* observed in children (see Wiesberg, 2015 for an overview). Symbolic pretense should be distinguished from behavioral imitation (see Adair & Carruthers, 2022 on the role of imitation in pretend play) in that the former involves what Nicols & Stich (2000, p. 119) call *inferential elaboration*: “Inference often plays a crucial role in filling out the details of what is happening in pretense.” Nicols & Stich (2000) suggest that symbolic pretense involves an initial premise, a suppositional state, within which the pretender “draws inferences about what is going on in the pretense.” In other words, pretense happens in a mental workspace, what Nicols & Stich call a *possible world box*, opened up by a suppositional state and the types of reasoning rules that apply inside the box are the same as those that operate on the actual beliefs. They discuss a three-year-old child, reported in Gould (1972), who, when climbing on a jungle gym, said: “I am a pussycat. Meow, meow”. After coming down from the gym, the boy laid on the ground and said: “I’m dead. I am a dead pussycat... I got shot.” Here the boy being a cat is the initial premise and him being dead after getting shot within this suppositional state involves types of reasoning that apply inside and outside of the “possible world box”. In (22), we provide a formal approximation of the mental processes underlying the child’s behavior.¹² Note that the unique cat is dead within the suppositional state in which the child is a pussycat. It follows, then, that, within this state, the child is pretend-dead.

¹² We remain agnostic as to whether symbolic pretense relies on the module for *theory of mind*, the capacity for mental state attribution (Leslie, 1987, 1994; Leslie & Roth, 1993). For arguments against mental state attribution in pretense, see Lillard (1993) and Currie (1998).

(22) The premise: [pussycat(the_self)]¹

	$\forall x. \text{get-shot}(x) \rightarrow \text{dead}(x)$ [World Knowledge]
	————— $\forall$ -E
$\text{get-shot}(\text{the_pussycat})$	$\text{get-shot}(\text{the_pussycat}) \rightarrow \text{dead}(\text{the_pussycat})$
	————— $\rightarrow$ E
	$\text{dead}(\text{the_pussycat})$

Are there convincing examples of symbolic pretense, the kind of pretense that involves inferential elaboration, in great apes? After some thirty years of natural observation at Bossou-Nimba (Guine), Matsuzawa (2020) reports of three instances of pretense in chimpanzee wild life. In one case, a young juvenile, by the name of Flanle, picked up a grass cushion used and left by a human and started walking bipedally while wearing it. This appears to be a case of behavioral imitation, where the chimpanzee “performed deferred imitation of a human model using the cushion to carry a heavy object on the head” (Matsuzawa, 2020, p. 546). In another episode, an 8.5-year-old female, Vuavua, carried a male hyrax killed by one of the other chimpanzees into her nest, where they stayed together for one night, and groomed him with her fingers and mouth, the way infant chimpanzees are typically groomed. The third episode of pretense reported in Matsuzawa (2020) features an 8-year-old female, Ja, who was observed to carry a log while Jire, her mother, was carrying her sick sister, then aged two and a half years. The log Ja was carrying on her shoulder was quite similar in size and shape to the sick infant that her mother was carrying. Gómez (2008) notes that such mothering-like behavior towards animate and inanimate object is prevalent in primate life. Young gorillas show such mothering-like behavior towards objects even in the absence of any adult model (see, Gómez & Martín-Andrade, 2002), which is reminiscent of clumsy caring-behavior associated with new gorilla mothers. We concur with Gómez’s (2008, p. 588) remark that “those rare instances of play mothering with objects observed in the wild need not in fact involve any pretend element.” Rather, they are indicative of “‘instinctive’ patterns whose function is to trigger actual maternal behavior with real babies.” In other words, such reports do not seem to require that we attribute apes the capacity for reasoning within a suppositional state of the kind we have seen in (22). We propose, as a hypothesis, that pretense, when it involves inferential elaboration, is unique to humans.

We have discussed the role that Conditional Introduction plays in various aspects of human cognition. How about Negation Introduction? There is evidence that humans apply this rule unconsciously in elimination of “nonfitting relations among cognitions” (Festinger, 1957, p. 3), where cognition, in this context, includes “any knowledge, opinion or belief about the environment, about oneself, or about one’s behavior” (p. 3). The presence of contradictory thoughts in cognition, where one of our thoughts entails the negation of another thought that we happen to entertain, leads to *cognitive dissonance*, which motivates an individual “to try to reduce the dissonance and achieve consonance” (Festinger, 1957, p.3, see also Cooper 2005, Quilty-Dunn, 2022). In one experiment (Festinger & Carlsmith, 1959), subjects were asked to express a positive emotion (i.e. it was interesting/intriguing/exciting etc.) to other people about a boring, monotonous task they had completed, a process called *forced compliance*. When asked about their honest opinion, the subjects who were paid one dollar expressed a more positive attitude towards the task than the subjects who were paid twenty dollars. Festinger & Carlsmith suggest the group that was compensated with little money employed a dissonance reducing strategy by updating their belief about the task. How exactly? Quilty-Dunn (2022) suggests that the generation of dissonance is a result of the following steps of reasoning: (1) The task is not worthwhile, (2) If the task is not worthwhile, then I am incompetent for freely choosing to engage in it and (3) I am incompetent for freely choosing to engage in this task. Below, we express this dissonance-inducing reasoning in the Gentzen form (“s” is the self and “t” is the task).

$$\begin{array}{c}
 (23) \quad \text{competent}(s) \rightarrow [\text{willingly.do}(s, t) \rightarrow \text{worthwhile}(t)] \quad \text{competent}(s) \quad [\text{by EGO}] \\
 \hline
 \text{willingly.do}(s, t) \rightarrow \text{worthwhile}(t) \quad \text{willingly.do}(s, t) \quad [\text{World Knowledge}] \\
 \hline
 \text{worthwhile}(t) \quad \neg (\text{worthwhile}(t)) \quad [\text{World Knowledge}] \\
 \hline
 \perp
 \end{array}$$

The participant has reached an *impasse*, a contradictory mental state, which is the source of cognitive dissonance. What must be done inside such a state is find a way to eliminate this

contradictory state. This can be achieved by turning one of the sentences in the reasoning into a supposition so that the contradictory thought can be relativized to this suppositional state. In this way, the contradictory state need not be the final state of the reasoning and, more importantly, the newly obtained suppositional sentence can be “blamed” for the rise of contradiction. For instance, the participant may give up on the idea of competence. That is not going to work, however.

Competence is one of those properties of our “self-concept” that we need to protect (Aronson, 1992), together with *Stability* and *Morality*. Instead the participant unconsciously turns the belief that “the task is not worthwhile” into a supposition. Moreover, since this supposition has led to a contradictory state, it must be false: It is *not* the case that the task is not worthwhile, from which it follows that the task is worthwhile (as “ $\neg\neg\phi$ ” entails “ ϕ ”).

$$\begin{array}{rcl}
 (24) & \text{competent}(s) \rightarrow [\text{willingly.do}(s, t) \rightarrow \text{worthwhile}(t)] & \text{competent}(s) \text{ [by EGO]} \\
 & \hline & \rightarrow E \\
 & \text{willingly.do}(s, t) \rightarrow \text{worthwhile}(t) & \text{willingly.do}(s, t) \text{ [World Knowledge]} \\
 & \hline & \rightarrow E \\
 & \text{worthwhile}(t) & \cancel{[\neg(\text{worthwhile}(t))]}^{\dagger} \\
 & \hline & \perp \\
 & \hline & \neg I^1 \\
 & \neg\neg(\text{worthwhile}(t)) & \\
 & \text{worthwhile}(t) &
 \end{array}$$

Reducing cognitive dissonance in a way that is sensitive to our self-concept, our ego, can be considered to be an instance of the psychological defense mechanism of Rationalization (Quilty-Dunn, 2022), which is possibly a component of our psychological immune system (Mandelbaum, 2019). The fact that such processes are handled at an unconscious level suggests that logic in general and the rule of Negation Introduction in particular are deeply woven into our mental life.

This concludes our discussion the implementations of indirect forms of reasoning in human cognition. In the next section, we shall take a closer look at the mental underpinnings of the capacity to manipulate rules of inference that have access to the history of a reasoning.

4. Procedural Memory and the grammar of looking back

There are important formal similarities between long-distance dependencies in syntax and suppositional states in reasoning. The representation of such processes requires access to earlier steps in a derivation; in other words, they are non-Markovian. In this section, we clarify the common core of these mental operations and discuss the role that an enhanced procedural memory might have played in these capacities.

Language consists at least of two subsystems: (a) the mental lexicon, which is the storage of basic units of computation, typically morphemes, and (b) grammar, which contains rules of symbol manipulation that are used in mental construction of hierarchically structured, semantically compositional complex expressions. Within Ullman's declarative/procedural model (the DP model, Ullman, 2004; Ullman & Pierpoint, 2005), this distinction between lexicon and grammar is more than a linguistic convention. It is reflected in the distinction between the declarative memory and the procedural memory, two memory systems supported by distinct brain circuitry. According to this model, the declarative memory system underlies the mental lexicon. "This system subserves the acquisition, representation and use not only of knowledge about facts and events but also words. It stores all arbitrary, idiosyncratic word-specific knowledge, including word meanings, word sounds, and abstract representations such as word categories." (Ullman, 2004, pp. 244 – 245). The procedural memory, which is taken to be a system enabling implicit (unconscious) acquisition of motor and cognitive capacities, underlies "the computation of already-learned, rule-based procedures that govern the regularities of language – particularly those procedures related to combining items into complex structures that have precedence (sequential) and hierarchical relations. Thus, this system is hypothesized to have an important role in rule-governed structure building" (Ullman, 2004, p. 245). Empirical evidence for the dissociation between lexicon and grammar in the brain comes from psycholinguistic (Ullman, 2001) and neuroimaging (Ullman, 2004) studies as well as from various forms of language impairment (see Ullman & Pierpoint, 2005 for an overview.)

In Minimalist syntax, Merge is the only operation responsible for building hierarchically-organized syntactic structure. This operation takes two syntactic objects, α and β , as input and defines the unique set that contains α and β as its sole members (Chomsky, 2013; Collins, 2002; Seely, 2006).

$$(25) \quad \text{Merge}(\alpha, \beta) = \{\alpha, \beta\}$$

When merging α and β , it may be that α is contained in β (i.e. $\beta = \{\dots\alpha\dots\}$, which can be read as “ α can be found somewhere within β ”) or it may be that it is not contained in β . If α is not contained in β , then this subcase of Merge is called external Merge (as α is *external* to β). If α is contained in β , then this is called internal Merge (as α is *in* β).¹³ Consider the derivation of the following sentences: (1) *There may be a student on the bench* and (2) *A student may be on the bench*. The relevant derivations start with merging elements of the quantificational nominal phrase and those of the prepositional phrase (Step 1, Step 2). These are all instances of external Merge. We, then, merge the nominal phrase with the preposition phrase to obtain some kind of predication structure (Step 3).

$$\text{Step 1. Merge}(a, \text{student}) = \{a, \text{student}\}$$

$$\text{Step 2. Merge}(\text{on}, \text{Merge}(\text{the}, \text{bench})) = \{\text{on}, \{\text{the}, \text{bench}\}\}$$

$$\text{Step 3. Merge}(\{a, \text{student}\}, \{\text{on}, \{\text{the}, \text{bench}\}\}) = \{\{a, \text{student}\}, \{\text{on}, \{\text{the}, \text{bench}\}\}\}$$

We add the verbal element *be* to the structure, as a result of which we get a verb phrase (Step 4). This verbal phrase is then merged with the modality operator *may* (Step 5), which is understood to carry the tense information as the present tense.

$$\text{Step 4. Merge}(\text{be}, \{\{a, \text{student}\}, \{\text{on}, \{\text{the}, \text{bench}\}\}\}) = \{\text{be}, \{\{a, \text{student}\}, \{\text{on}, \{\text{the}, \text{bench}\}\}\}\}$$

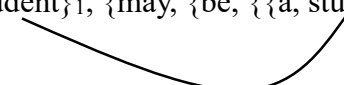
$$\text{Step 5. Merge}(\text{may}, \{\text{be}, \{\{a, \text{student}\}, \{\text{on}, \{\text{the}, \text{bench}\}\}\}\}) = \{\text{may}, \{\text{be}, \{\{a, \text{student}\}, \{\text{on}, \{\text{the}, \text{bench}\}\}\}\}\}$$

¹³ Historically, the operation of internal Merge was considered to be as an instantiation of movement transformation.

The English grammar has the requirement that a functional morpheme that carries the tense information (here, the tense head *may*) should be merged with a nominal element (This is sometimes called the Extended Projection Principle (EPP) property, Chomsky, 1981; 1995). We can think of this requirement as the tense head carrying a feature, the EPP feature, that must be checked via merger with a nominal phrase. There are now two ways to satisfy this requirement. We can *externally* merge ‘there’ to the output of Step 5 to obtain the sentence “There may be a student on the bench” (Step 6a). Or else, we may search into the phrase obtained in Step 5 and find a nominal phrase within it that can satisfy the merge requirement on the tense head. After scanning this syntactic representation, we detect the embedded quantificational expression $\{a, student\}$ inside it and re-merge this nominal phrase with the structure obtained in Step 5: “ $\{a, student\}_1$ may be [~~$\{a, student\}_1$~~ on the bench]” (Step 6b). This is an instance of internal Merge. The strikethrough on the lower copy indicates that it is not to be pronounced at the level of externalization. To express the identity of the phrase $\{a, student\}$ in both positions, we index them with the same number.

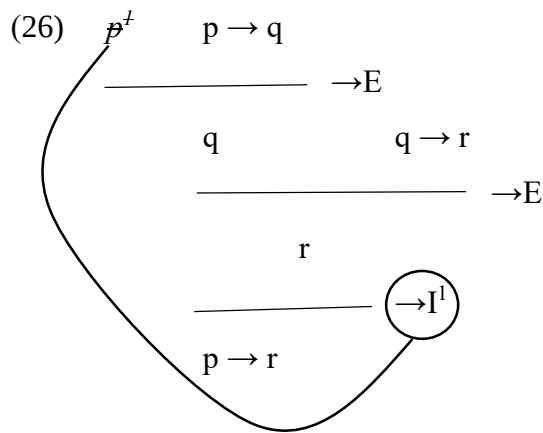
Step 6a. Merge(there, {may, {be, {{a, student}, {on, {the, bench}}}}})) (External)

Step 6b. Merge($\{a, student\}_1$, {may, {be, {{a, student}_1, {on, {the, bench}}}}})) (Internal)



As Piattelli-Palmarini & Uriagereka (2005) note, a grammar that can do internal Merge is a grammar that allows access to derivation history (as in Step 6b). This capacity puts a high burden on the procedural memory (Bolender et al., 2008). In fact, the kind of procedural memory that can implement rules that make note of steps taken earlier in the derivation is likely to be uniquely human (Piattelli-Palmarini & Uriagereka, 2005, Bolender et al. 2008). In other words, a boosted procedural memory that is operative in mental manipulation of rules that can access derivational history has been claimed to be *specifically human*, as Piattelli-Palmarini & Uriagereka (2005) put it. Interestingly, the contrast between external Merge and internal Merge in terms of derivational memory is reminiscent of the contrast between direct and indirect forms of reasoning. Just like internal Merge, indirect forms of reasoning require access to earlier steps in the proof, more specifically access to the sentence under supposition. This is exemplified below in the reasoning (26), repeated from (7). While the rule of Conditional Elimination, which is an

instance of a direct form of reasoning, needs no access to earlier steps in the derivation, the rule of Conditional Introduction targets the suppositional sentence introduced earlier in the proof.



If, as Piattelli-Palmarini & Uriagereka (2005) and Bolender et al. (2008) suggest, the uniqueness of internal Merge to human grammar is to be explained by the evolution of enhanced procedural memory within the human lineage, then we also get an immediate account of the uniqueness of indirect forms of reasoning to human mind. As Bolender et al. (2008, p. 143) note, the cognitive functions enabled by an enhanced procedural memory “may have played an important role in the great computational power of syntax in language,” which, consequently, would explain “the qualitatively different computational, and ultimately conceptual, powers of the human mind.” We shall also add that enhanced procedural memory may also underlie suppositional forms of reasoning, which are, by our hypothesis, unique to humans.

5. Conclusion

The distinction between suppositional and non-suppositional forms of reasoning is a logical one. If this paper is on the right track, it is also psychologically real. It is the distinction between the forms of reasoning that are unique to humans and those that are possibly quite widespread in the animal kingdom. This suggests, more generally, that it may be a worthwhile effort to re-analyze logical concepts in mental terms and study the consequences of such a reinterpretation.

Underlying this paper is the idea that a detailed understanding of logical structures, be they classical or non-classical (Restall, 2000; Priest, 2008), should be a crucial component of the study of animal cognition in general and human cognition in particular.

In short, logic is more than a subject in discrete mathematics. It is a mind-internal capacity that we make use of in our dealings with the external world. The unique way in which we experience the world is, in part, due to the unique properties of the logic that we internalize as members of *homo sapiens*.

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